



Llava-Med 7B

660k image text pairs across multiple modalities
Medical Expert – specialized curriculum

- Visual Question Answering
- Classification and Captioning
- Grounding



Is scale everything?

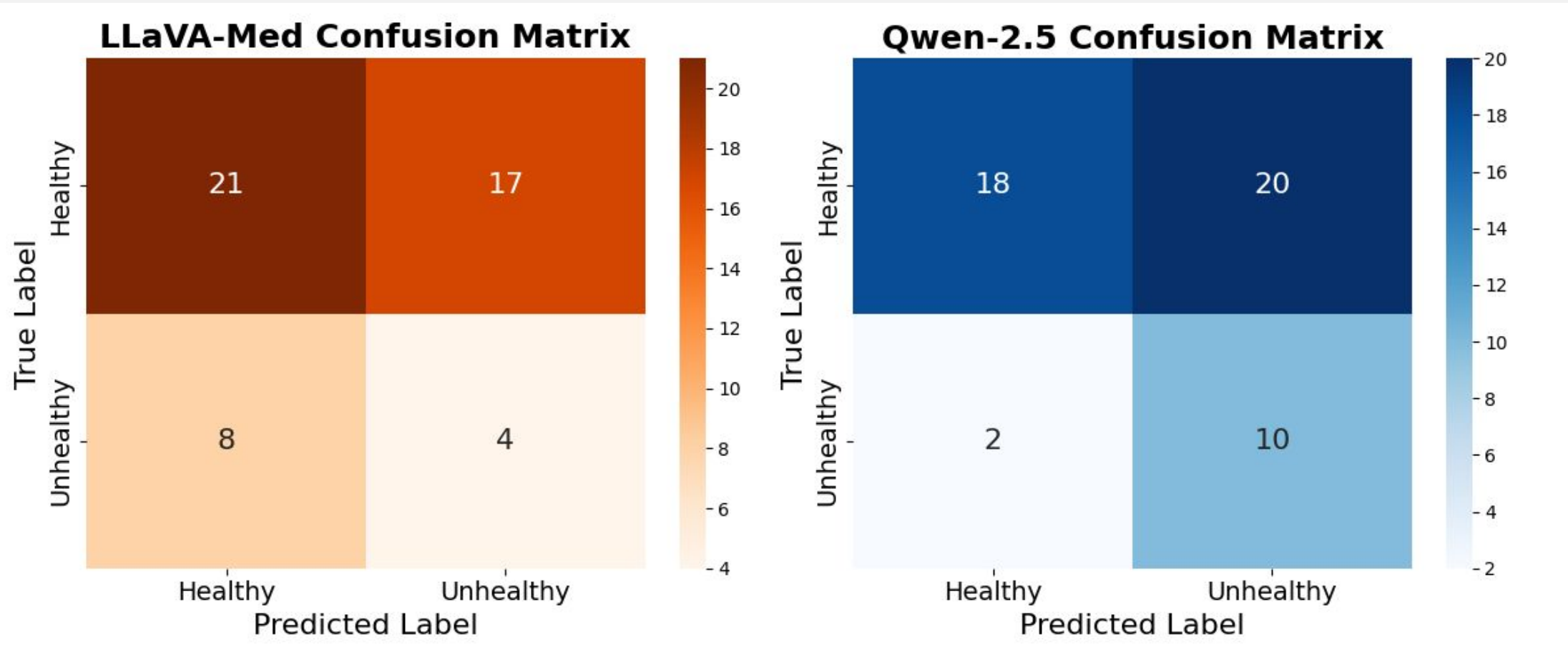
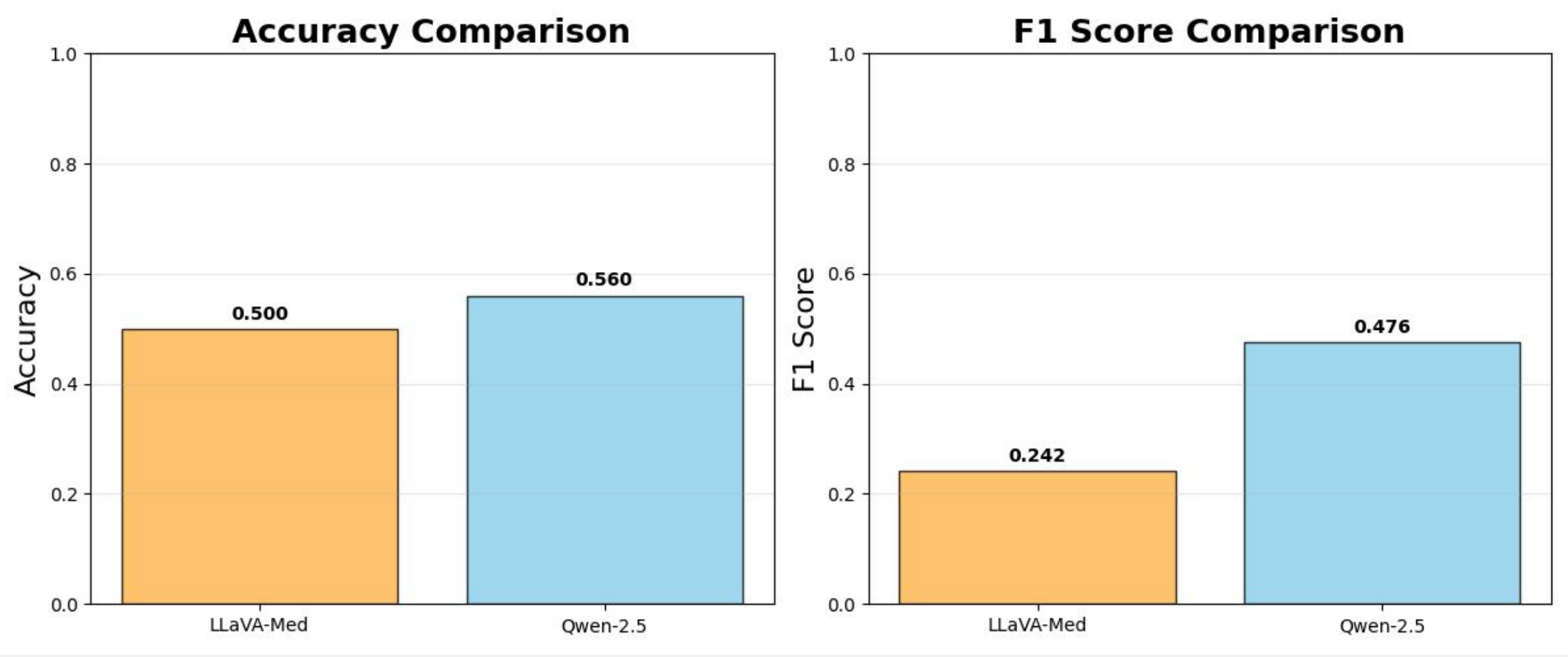
4 Trillion Tokens of high-quality data
Generalist foundation model – optimized for robust performance

- Visual Question Answering
- Classification and Captioning
- Grounding



Qwen2.5-VL 72B

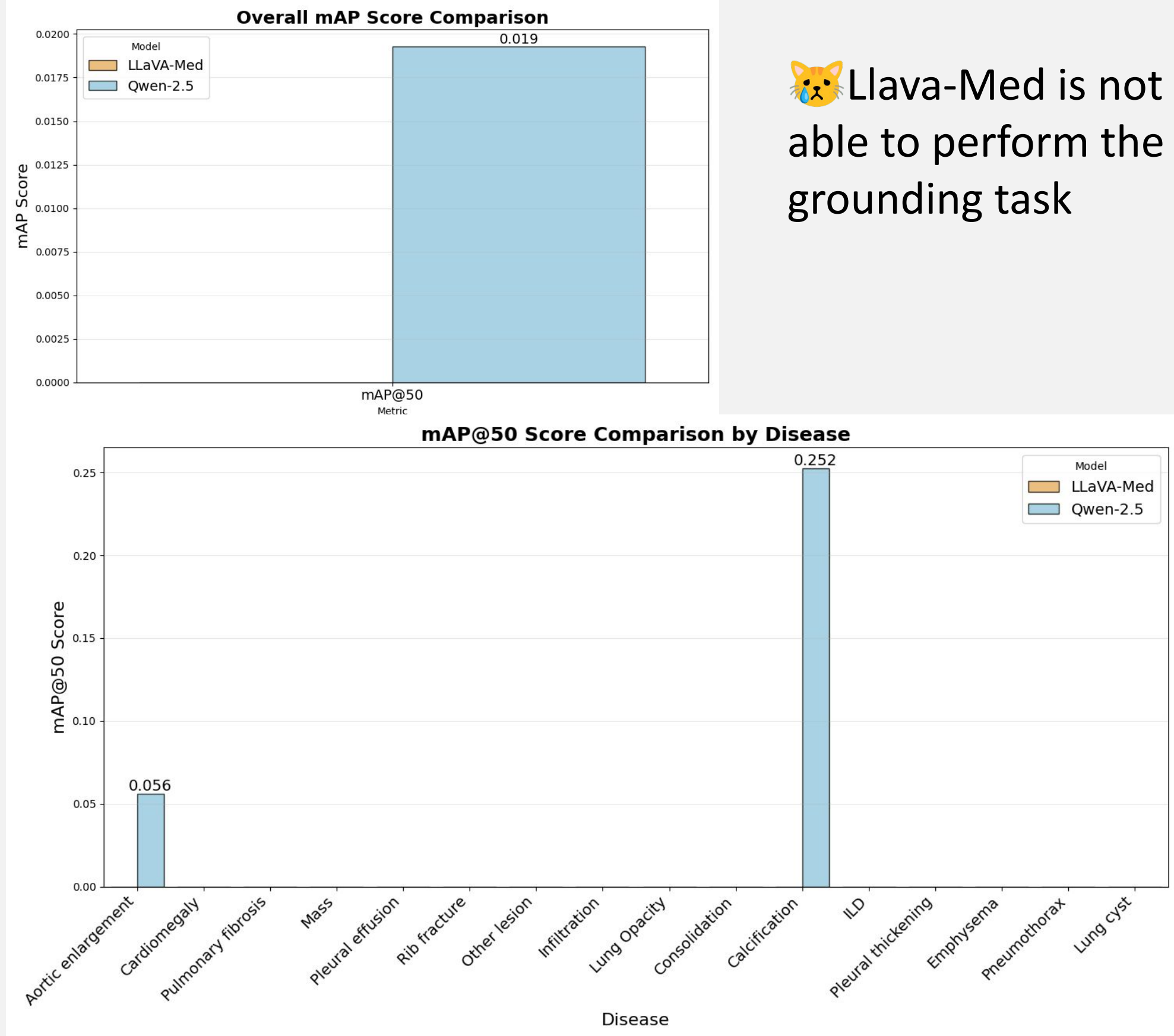
Chest X-ray Classification



Evaluation:
2 Modalities
5 Tasks

Generalist models
outperform smaller
domain experts

Chest X-ray Grounding



LLaVA-Med is not able to perform the grounding task

Medical Domain
Tasks are challenging
for both models

Brain MRI Description

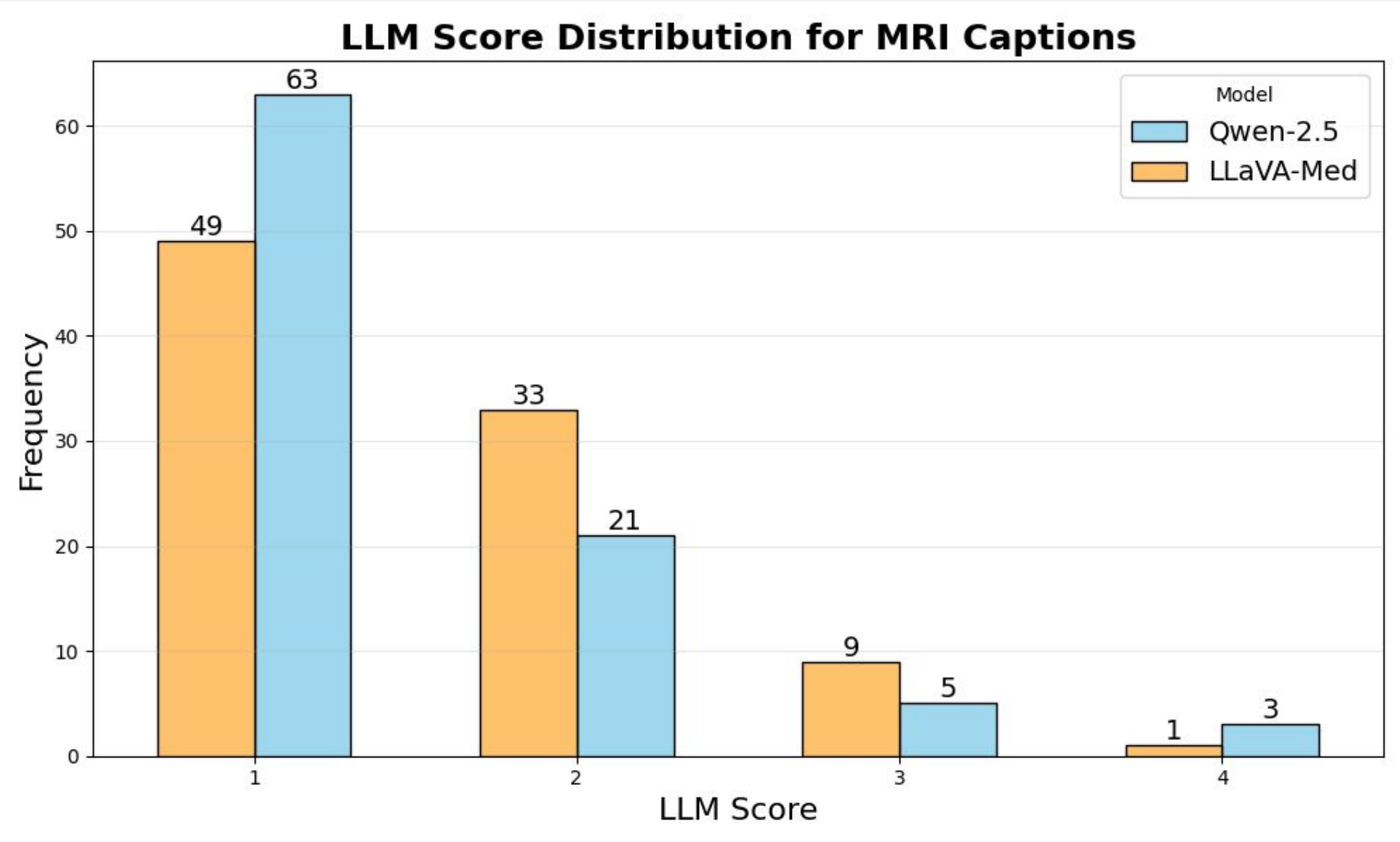
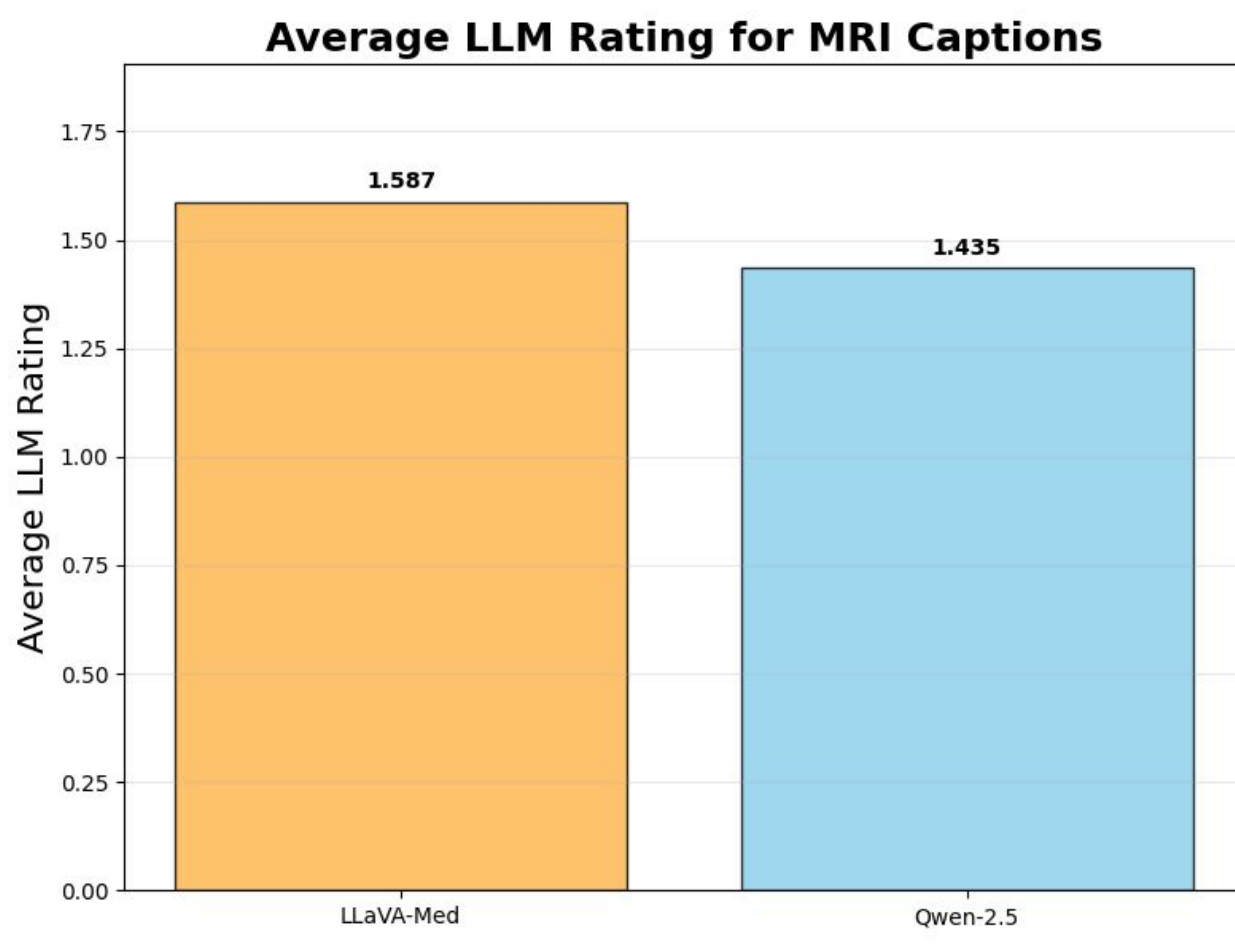


Table 1: BLEU Score Comparison between LLaVA-Med and Qwen-2.5 for MRI Caption Generation

Metric	LLaVA-Med	Qwen-2.5
BLEU-1	0.1034 ± 0.0656	0.0967 ± 0.0724
BLEU-2	0.0273 ± 0.0345	0.0309 ± 0.0328
BLEU-3	0.0030 ± 0.0126	0.0046 ± 0.0171
BLEU-4	0.0000 ± 0.0000	0.0000 ± 0.0000

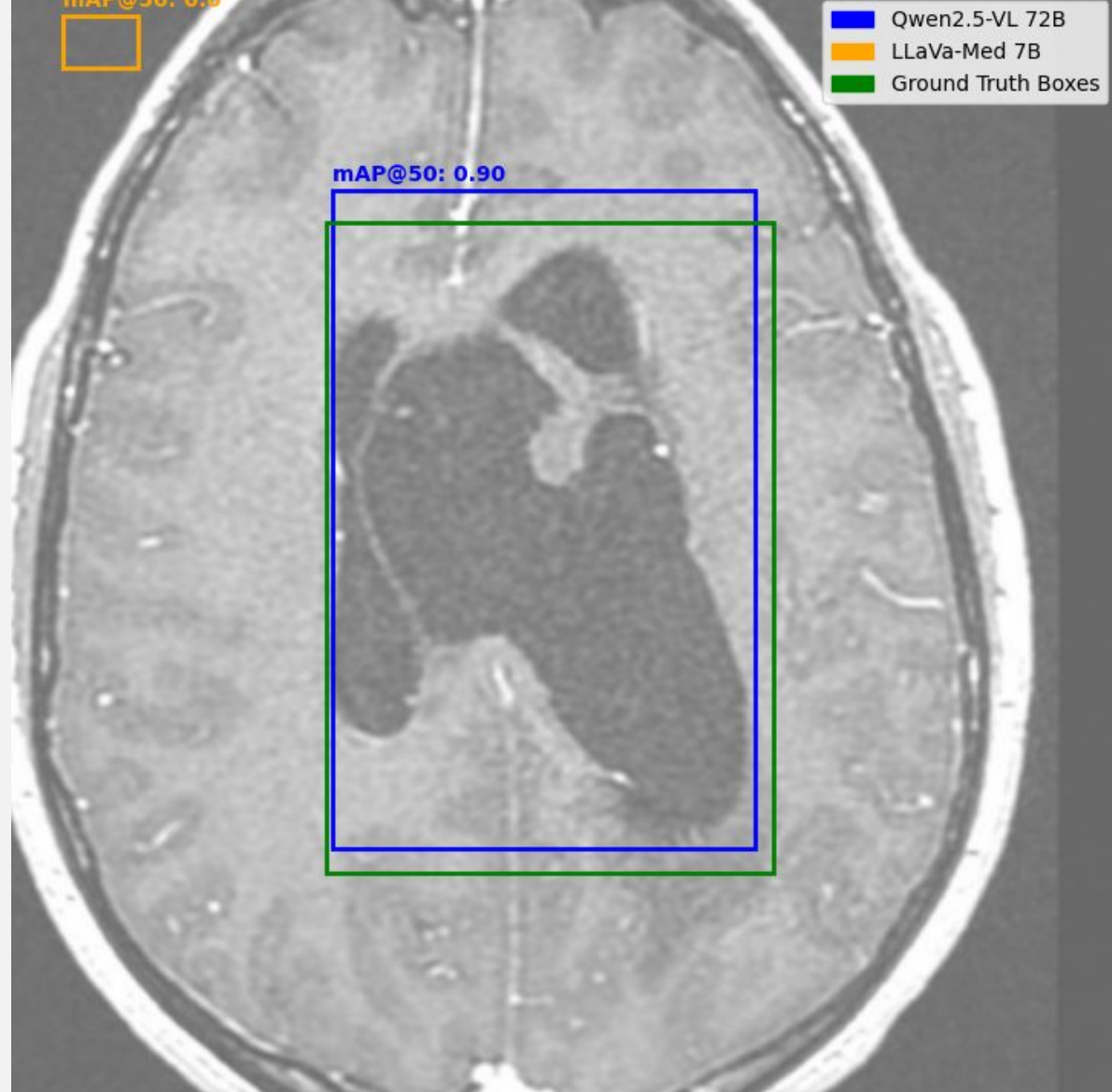
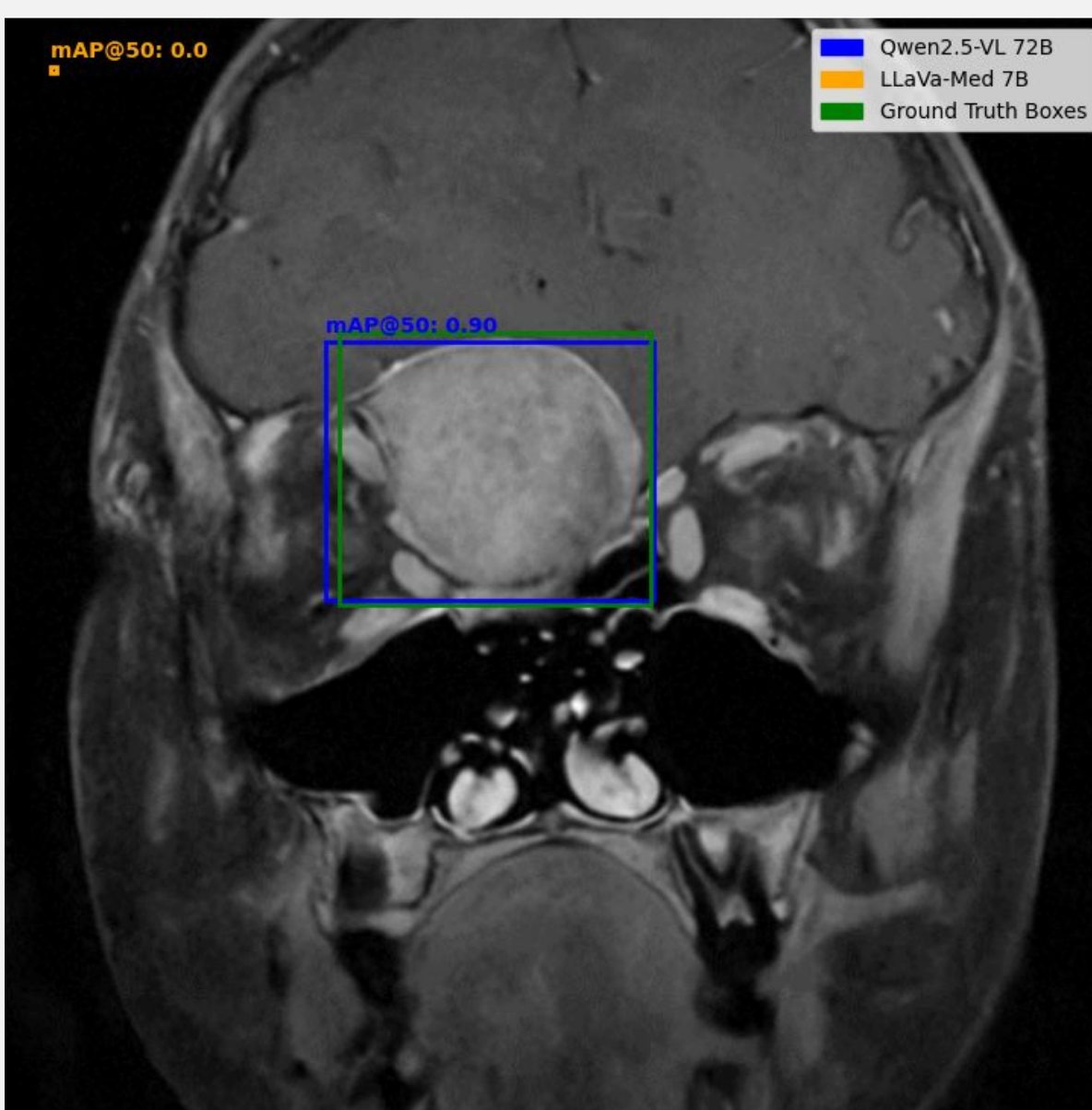
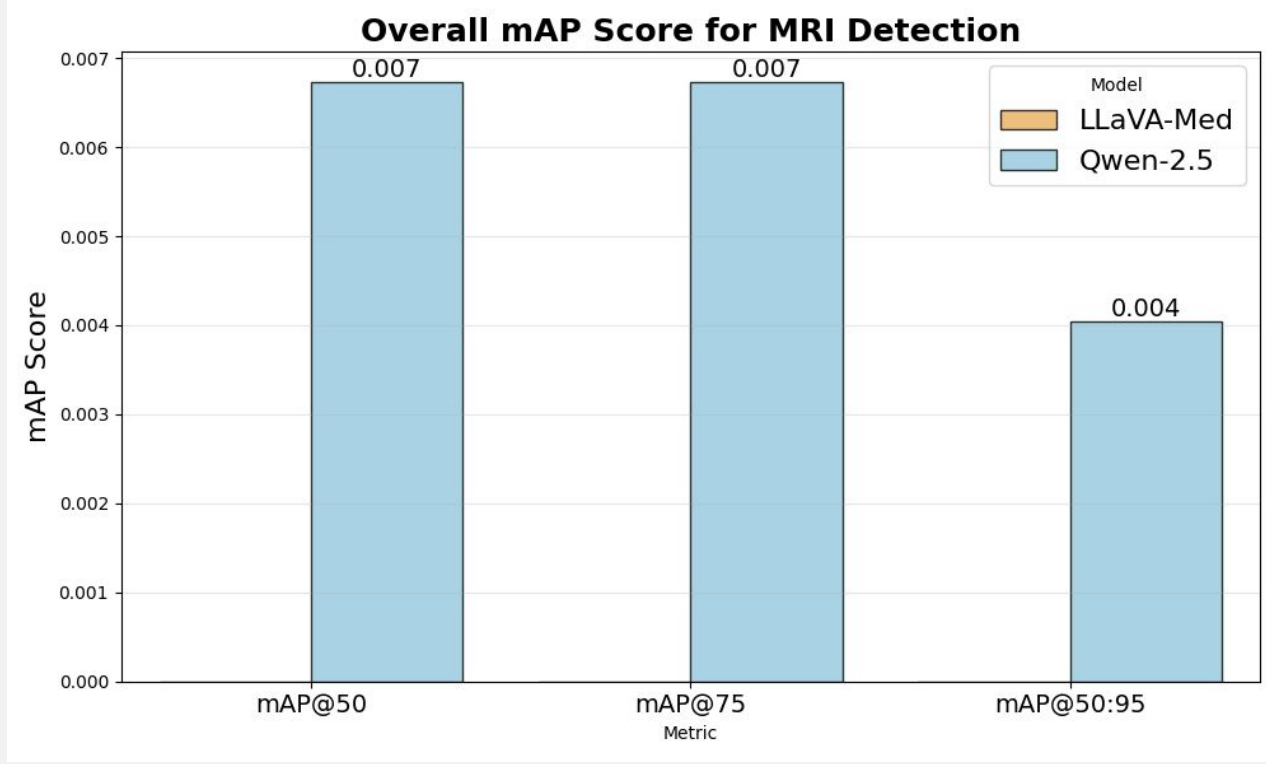
LLM Score - LLM as a Judge
BLEU Score - based on n-grams



Brain MRI Detection

Better performance compared to the chest grounding task

As expected Llava-Med fails completely



Brain MRI Diagnosis

Given the medical history and corresponding brain slices diagnose the patient:

Llava-Med: 2 / 25 correct
Qwen2.5-VL: 3 / 25 correct

Still way off from an acceptable zero-shot performance

